where M is the mass of the Earth and V its volume, Thus,

$$\rho = \frac{M'}{V'}$$

$$\frac{M'}{V'} = \frac{M}{V} \text{ and } M' = \frac{M}{V}V'$$

$$M' = \left(\frac{M}{\frac{4}{3}\pi R_e^{\,3}}\right)\left(\frac{4}{3}\pi(R_e - d)^3\right)$$

$$M' = \frac{M}{R_e^{\,3}}(R_e - d)^3 \qquad (6.49)$$

Substituting equation (6.49) in equation (6.47)

$$g' = G\frac{M}{R_e^{\,3}}(R_e - d)^3 . \frac{1}{(R_e - d)^2}$$

$$g' = GM\frac{R_e\left(1 - \frac{d}{R_e}\right)}{R_e^{\,3}}$$

$$g' = GM\frac{\left(1 - \frac{d}{R_e}\right)}{R_e^{\,2}}$$

Thus

$$g' = g\left(1 - \frac{d}{R_e}\right) \qquad (6.50)$$

Here also $g' < g$. As depth increases, g' decreases. It is very interesting to know that acceleration due to gravity is maximum on the surface of the Earth but decreases when we go either upward or downward.

Variation of g with latitude:

Whenever we analyze the motion of objects in rotating frames [explained in chapter 3] we must take into account the centrifugal force. Even though we treat the Earth as an inertial frame, it is not exactly correct because the Earth spins about its own axis. So when an object is on the surface of the Earth, it experiences a centrifugal force that depends on the latitude of the object on Earth. If the Earth were not spinning, the force on the object would have been mg. However, the object experiences an additional centrifugal force due to spinning of the Earth.

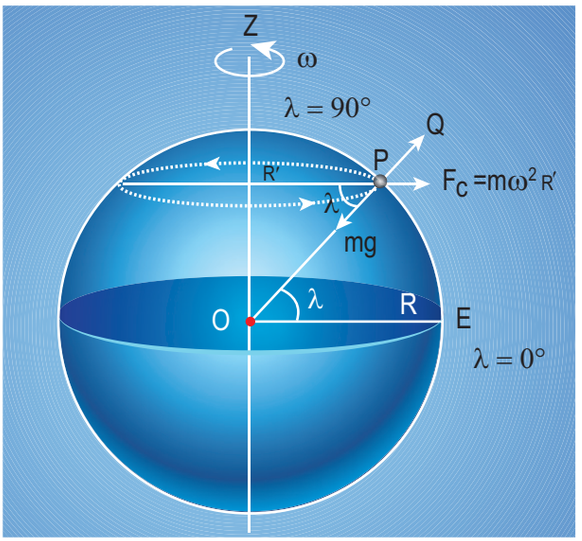


Figure 6.18 Variation of g with latitude

This centrifugal force is given by $m\omega^2 R'$.

$$R' = R\cos\lambda \qquad (6.51)$$

where λ is the latitude. The component of centrifugal acceleration experienced by the object in the direction opposite to g is

$$a_{PQ} = \omega^2 R'\cos\lambda = \omega^2 R\cos^2\lambda$$

$$\text{since } R' = R\cos\lambda$$

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